

Econ 646 Spring 2011
Homework 5

The due date for this assignment is April 5th at the beginning of the tutorial.

Question 1 (household's problem): A representative household's utility function is

$$U(C)$$

with $U'(\cdot) > 0$. C is a consumption index given by

$$C = \left[\int_0^N C(i)^{\frac{\varepsilon-1}{\varepsilon}} di \right]^{\frac{\varepsilon}{\varepsilon-1}}$$

with $C(i)$ representing the quantity of good i consumed by the household. I assume the existence of a continuum of goods represented by $[0, N]$ interval. ε is greater than 1. H hours of time are inelastically supplied for work (in other words, H is fixed, not a choice variable). The budget constraint is

$$\int_0^N P(i) C(i) di = WH + T$$

where W is the nominal wage and T is a lump-sum component of income, or more specifically, T is the dividend income (the household is the owner of the firms). Since the right-hand side of the budget constraint is exogenous to the household, the remaining problem is finding the optimal allocation of consumption expenditure, which we will solve below.

1. The problem of maximizing C given

$$\int_0^N P(i) C(i) di = WH + T \tag{1}$$

can be expressed by means of the Lagrangean

$$\mathcal{L} = \left[\int_0^N C(i)^{\frac{\varepsilon-1}{\varepsilon}} di \right]^{\frac{\varepsilon}{\varepsilon-1}} - \lambda \left[\int_0^N P(i) C(i) di - WH - T \right]$$

Derive the associated first-order conditions

$$\frac{P(i)}{P(j)} = \left(\frac{C(i)}{C(j)} \right)^{-\frac{1}{\varepsilon}}$$

for any $i, j \in [0, N]$.

2. The first order condition can be written as

$$C(i) = C(j) \left(\frac{P(i)}{P(j)} \right)^{-\varepsilon} \tag{2}$$

Show that substituting (2) into (1) obtains

$$C(j) \left(\frac{1}{P(j)} \right)^{-\varepsilon} \int_0^N P(i)^{1-\varepsilon} di = WH + T \tag{3}$$

3. Define $P = \left[\int_0^N P(i)^{1-\varepsilon} di \right]^{\frac{1}{1-\varepsilon}}$. Show that (3) can be written as

$$C(j) = \left(\frac{P(j)}{P} \right)^{-\varepsilon} \frac{WH + T}{P} \quad (4)$$

4. By substituting (4) into the definition of C , show that

$$\int_0^N P(i) C(i) di = PC$$

and therefore the total expenditure is the product of price index times the quantity index.

5. (4) is the downward sloping demand curve. Draw two demand curves with different values of ε in a diagram with $P(j)$ plotted on the vertical axis and $C(j)$ plotted on the horizontal axis. Clearly indicate which curve corresponds to a high ε and which curve corresponds to a low ε .

Question 2 (firm's problem): Each firm produces a differentiated good with technology

$$Y(i) = H(i)$$

$H(i)$ is firm i 's labor input. There is no price stickiness. Each firm chooses $P(i)$ to maximize profits defined as

$$P(i) Y(i) - WH(i)$$

subject to the demand constraint

$$Y(i) = \left(\frac{P(i)}{P} \right)^{-\varepsilon} C$$

Individual firms take aggregate price level P and aggregate consumption index C as given.

1. Show that the optimal price is

$$P(i) = \frac{\varepsilon}{\varepsilon - 1} W$$

(HINT: profits can be written as $(P(i) - W) Y(i) = (P(i) - W) \left(\frac{P(i)}{P} \right)^{-\varepsilon} C$.)

2. Let $\pi(i)$ denote the maximized profits. What is the ratio of profits to costs, i.e., $\pi(i) / (WH(i))$, for the firm which is optimally choosing $P(i)$? How does it depend on the parameter ε ? Why does it so?

Question 3 (equilibrium): Goods market clearing condition and labor market clearing condition require

$$C(i) = Y(i) \text{ for any } i \in [0, N]$$

$$H = \int_0^N H(i) di$$

Because of the symmetry between firms, we can easily find $H(i) = H/N$ in equilibrium.

1. Find the equilibrium aggregate consumption index C .
2. Find the equilibrium relative price $P(i)/P$ for any $i \in [0, N]$.
3. Find the equilibrium real wage W/P .
4. Find the labor share WH/PC and the profits share $\pi(i)N/PC$ in equilibrium.
5. Is the equilibrium allocation efficient? Answer yes or no, and explain why.
6. Let's do a simple comparative statics with respect to N . How do C , $P(i)/P$, and W/P change when N gets larger? Give intuition for each of those.